

IMO 2026 — Problem 2

Solution by Claude Fable 5

2026-07-17

Source: `results/imo-2026/claude-fable-5/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC , respectively. Let points K and L be chosen inside triangles BMC and BNC , respectively, such that K lies inside the angle LBA , L lies inside the angle ACK , and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL . Prove that $OM = ON$.

Solution document

Status

solved

Approaches tried

- **complex-certificate — worked (APPROVED, round 1).** Branch-pinned law-of-sines encoding of all five hypothesis conditions into two polynomial side relations $_K = 0$, $_L = 0$, plus an unconditional trigonometric certificate $2 \sin A \cdot G = _K + _L$. Every displayed identity and every Fourier-table row independently re-verified symbolically (sympy, exp-rewrite) and the whole chain re-verified end-to-end on independently constructed configurations by the reviewer (`/tmp/round-1/review/`). Non-collinearity of A, K, L proved from the hypotheses (Part 4).
- **secant-trig-identity — worked (APPROVED, round 1).** Independent second complete proof: same reduction skeleton, constraints $(K)/(L)$ equivalent to $_K/_L$, closed by the four-variable Key Identity $N(s)U + N(t)V + W = \sin(s-t)[N(s)N(t) - \sin^2s \sin^2t]$, proved by a degree-3 interpolation argument at $s \in \{t, 0, \pi, -\pi\}$ plus density/continuity. All steps re-verified symbolically by the reviewer. (Uses the problem statement’s “triangle AKL ” for non-collinearity, flagged explicitly; complex-certificate proves that fact independently.)
- **midpoint-reflection-isogonal — partial.** Correct reflection dictionary ($K^* = 2M-K$, $L^* = 2N-L$, exterior isogonal rays at A) and a rigorous directed-length reduction $OM = ON$ (I): $BK \cdot AP + AK^2 - CL \cdot AP - AL^2 = (c^2 - b^2)/2$ (Lemmas 1–4, reviewer-verified). The

synthetic derivation of (I) from the transferred angle hypotheses remains open; extensive candidate list refuted. Superseded by the two solved approaches.

Current best

Problem solved. Canonical proof below (from `approaches/complex-certificate.md`); an independent second complete proof is in `approaches/secant-trig-identity.md`.

Full proof

Problem. Let ABC be a triangle, M, N the midpoints of AB, AC . Points K, L are chosen inside triangles BMC, BNC respectively, such that K lies inside angle LBA , L lies inside angle ACK , and

$$\angle KBA = \angle ACL, \quad \angle LBK = \angle LNC, \quad \angle LCK = \angle BMK.$$

Let O be the circumcentre of triangle AKL . Prove $OM = ON$.

Throughout, $b = CA$, $c = AB$, A also denotes the angle $\angle BAC \in (0, \pi)$, and all named angles are **unsigned angles in** $(0, \pi)$ (as in the problem statement). Set

$$\begin{aligned} \varphi &:= \angle KBA = \angle ACL, & \psi &:= \angle LBK = \angle LNC, & \chi &:= \angle LCK = \angle BMK, \\ u &:= \angle KAB, & v &:= \angle LAC. \end{aligned}$$

The proof has six parts:

- **Part 0** — configuration facts (all interiority hypotheses are consumed here);
- **Part 1** — the unconditional reduction $OM = ON \iff OB^2 - OC^2 = \frac{c^2 - b^2}{2}$;
- **Part 2** — the law-of-sines encoding of the hypotheses;
- **Part 3** — elimination of χ and ψ : the two side-relations $\ell_K = 0, \ell_L = 0$;
- **Part 4** — A, K, L are not collinear;
- **Part 5** — coordinates, the circumcentre, and the reduction of the goal to one trig polynomial $G = 0$;
- **Part 6** — the certificate: the identity $2 \sin A \cdot G = \alpha \ell_K + \beta \ell_L$, verified in full.

Part 0. Configuration facts

(0a) K, L lie in the interior of triangle ABC . Since B, M, C lie in the (convex) closed triangle ABC , the closed triangle BMC is contained in the closed triangle ABC . The boundary of ABC is contained in the three lines AB, BC, CA . The open triangle $\text{int}(BMC)$ is disjoint from line AB (which contains its edge BM : the interior of a triangle lies strictly on one side of each edge line) and from line BC (its edge BC). Finally, the closed triangle BMC meets line CA only in the single point C : the vertices B and M lie strictly on the same side of line CA (every point of segment AB other than A does), so by convexity any point of triangle BMC on line CA lies on the edge... more directly: a point of the convex hull of $\{B, M, C\}$ lying on line CA must have zero total weight on B and M (their signed distances to the line have the same strict sign), hence equals C . Since the vertex C is not an interior point, $\text{int}(BMC)$ is disjoint from all three lines, hence from $\partial(ABC)$, so $\text{int}(BMC) \subseteq \text{int}(ABC)$. Thus $K \in \text{int}(ABC)$. The same argument with

N in place of M (edge NC on line CA , edge BC on line BC , and triangle BNC meeting line AB only at B) gives $\text{int}(BNC) \subseteq \text{int}(ABC)$, so $L \in \text{int}(ABC)$.

(0b) Basic angle facts. Since $K \in \text{int}(ABC)$, K lies on none of the lines AB , BC , CA ; likewise L . By the crossbar theorem, for an interior point K the ray AK meets segment BC , hence lies strictly inside angle BAC , so

$$\angle BAC = \angle BAK + \angle KAC, \quad \text{i.e.} \quad 0 < u < A, \quad \angle KAC = A - u > 0,$$

and likewise $0 < v < A$, $\angle LAB = A - v$. Similarly $0 < \varphi = \angle KBA < \angle B$ (interior point seen from vertex B) and $0 < \varphi = \angle ACL < \angle C$; in particular $\varphi \in (0, \pi)$ and $\sin \varphi > 0$. Since $K \notin \text{line } AB$ and $M \in \text{line } AB$ with $M \neq K$, the angle $\chi = \angle BMK$ is a genuine angle in $(0, \pi)$, so $\sin \chi > 0$ and $\chi > 0$; likewise $\psi = \angle LNC \in (0, \pi)$, $\psi > 0$.

(0c) Ray identifications at the midpoints. M is the midpoint of AB , so M lies strictly between B and A ; hence ray $BM = \text{ray } BA$ and $\angle KBM = \angle KBA = \varphi$. Likewise N lies strictly between C and A , so ray $CN = \text{ray } CA$ and $\angle LCN = \angle LCA = \varphi$.

(0d) Angle additivity from the two interiority hypotheses. “ K lies inside angle LBA ” means ray BK lies strictly inside angle LBA , so

$$\angle ABL = \angle ABK + \angle KBL = \varphi + \psi.$$

“ L lies inside angle ACK ” means ray CL lies strictly inside angle ACK , so

$$\angle ACK = \angle ACL + \angle LCK = \varphi + \chi.$$

These two lines and the positivity facts of (0b) are the only places the interiority hypotheses are used; everything after this point is unconditional algebra on genuine triangles.

Part 1. Reduction: $OM = ON \iff OB^2 - OC^2 = \frac{c^2 - b^2}{2}$

Median-length formula. For any points O, P, Q with S the midpoint of PQ : writing vectors from O , $4OS^2 = |\vec{OP} + \vec{OQ}|^2 = OP^2 + OQ^2 + 2\vec{OP} \cdot \vec{OQ}$ and $PQ^2 = OP^2 + OQ^2 - 2\vec{OP} \cdot \vec{OQ}$; adding, $4OS^2 = 2OP^2 + 2OQ^2 - PQ^2$.

Applying this to $S = M$ (midpoint of AB) and $S = N$ (midpoint of AC):

$$4OM^2 = 2OA^2 + 2OB^2 - c^2, \quad 4ON^2 = 2OA^2 + 2OC^2 - b^2.$$

Subtracting, $4(OM^2 - ON^2) = 2(OB^2 - OC^2) - (c^2 - b^2)$. Hence

$$OM = ON \iff OB^2 - OC^2 = \frac{c^2 - b^2}{2}. \quad (1)$$

(This step is unconditional; it does not even use that O is the circumcentre.)

Part 2. The hypotheses as law-of-sines relations

Each relation below is the law of sines in a genuine (nondegenerate) triangle; by (0b), $K \notin \text{lines } AB, CA$ and $L \notin \text{lines } AB, CA$, so all six triangles used are nondegenerate, all their angles lie in $(0, \pi)$, and every sine below is **positive**. When a triangle has angles θ_1, θ_2 at two vertices, the sine of the third angle is $\sin(\pi - \theta_1 - \theta_2) = \sin(\theta_1 + \theta_2)$.

Triangle ABK (angles u at A , φ at B):

$$AK = \frac{c \sin \varphi}{\sin(\varphi + u)}, \quad BK = \frac{c \sin u}{\sin(\varphi + u)}. \quad (E1)$$

Triangle MBK (angles φ at B by (0c), χ at M ; $MB = c/2$):

$$BK = \frac{(c/2) \sin \chi}{\sin(\varphi + \chi)}. \quad (E3')$$

Equating the two expressions for BK and clearing the (positive) denominators:

$$2 \sin u \sin(\varphi + \chi) = \sin \chi \sin(\varphi + u). \quad (E3)$$

Triangle ACK (angles $A - u$ at A by (0b), $\varphi + \chi$ at C by (0d)):

$$AK = \frac{b \sin(\varphi + \chi)}{\sin(\varphi + \chi + A - u)}. \quad (E5)$$

Triangle ACL (angles v at A , φ at C):

$$AL = \frac{b \sin \varphi}{\sin(\varphi + v)}, \quad CL = \frac{b \sin v}{\sin(\varphi + v)}. \quad (E2)$$

Triangle NCL (angles φ at C by (0c), ψ at N ; $NC = b/2$): equating with (E2)'s CL :

$$2 \sin v \sin(\varphi + \psi) = \sin \psi \sin(\varphi + v). \quad (E4)$$

Triangle ABL (angles $A - v$ at A , $\varphi + \psi$ at B by (0d)):

$$AL = \frac{c \sin(\varphi + \psi)}{\sin(\varphi + \psi + A - v)}. \quad (E6)$$

Part 3. Eliminating χ and ψ : the side relations

K-side. From (E3), expanding $\sin(\varphi + \chi) = \sin \varphi \cos \chi + \cos \varphi \sin \chi$ and using $\sin(\varphi + u) - 2 \sin u \cos \varphi = \sin \varphi \cos u - \cos \varphi \sin u = \sin(\varphi - u)$:

$$\sin \chi \sin(\varphi - u) = 2 \sin u \sin \varphi \cos \chi.$$

Since $\sin \chi > 0$ and $2 \sin u \sin \varphi > 0$ (Part 0), we may write

$$(\cos \chi, \sin \chi) = r (\sin(\varphi - u), 2 \sin u \sin \varphi), \quad r := \frac{\sin \chi}{2 \sin u \sin \varphi} > 0. \quad (2)$$

(No branch ambiguity: $\sin \chi > 0$ pins the sign of r , and (2) reproduces both $\cos \chi$ and $\sin \chi$ exactly.)
Consequently

$$\sin(\varphi + \chi) = \sin \varphi \cos \chi + \cos \varphi \sin \chi = r \sin \varphi [\sin(\varphi - u) + 2 \sin u \cos \varphi] = r \sin \varphi \sin(\varphi + u), \quad (3)$$

$$\cos(\varphi + \chi) = \cos \varphi \cos \chi - \sin \varphi \sin \chi = r [\cos \varphi \sin(\varphi - u) - 2 \sin u \sin^2 \varphi]. \quad (4)$$

Now equate (E1) and (E5) (both equal AK) and clear denominators:

$$c \sin \varphi \sin(\varphi + \chi + A - u) = b \sin(\varphi + \chi) \sin(\varphi + u).$$

Expand $\sin(\varphi + \chi + A - u) = \sin(\varphi + \chi) \cos(A - u) + \cos(\varphi + \chi) \sin(A - u)$ and substitute (3), (4); every term acquires the factor r , and dividing by $r \sin \varphi > 0$ leaves

$$c [\sin \varphi \sin(\varphi + u) \cos(A - u) + \cos \varphi \sin(\varphi - u) \sin(A - u) - 2 \sin^2 \varphi \sin u \sin(A - u)] = b \sin^2(\varphi + u). \quad (5)$$

Bracket identity. The bracket in (5) equals $\cos A - \cos(\varphi + u) \cos(A + \varphi - u)$. *Verification by product-to-sum (each row is the standard expansion $2 \sin X \sin Y = \cos(X - Y) - \cos(X + Y)$, $2 \sin X \cos Y = \sin(X + Y) + \sin(X - Y)$, $2 \cos X \cos Y = \cos(X - Y) + \cos(X + Y)$, applied twice); we list the five summands of (bracket) $- [\cos A - \cos(\varphi + u) \cos(A + \varphi - u)]$ in Fourier form:*

term	Fourier expansion
$\sin \varphi \sin(\varphi + u) \cos(A - u)$	$\frac{1}{4} \cos A - \frac{1}{4} \cos(A + 2\varphi) + \frac{1}{4} \cos(A - 2u) - \frac{1}{4} \cos(2\varphi + 2u - A)$
$\cos \varphi \sin(\varphi - u) \sin(A - u)$	$\frac{1}{4} \cos A + \frac{1}{4} \cos(A - 2\varphi) - \frac{1}{4} \cos(A - 2u) - \frac{1}{4} \cos(A + 2\varphi - 2u)$
$-2 \sin^2 \varphi \sin u \sin(A - u)$	$\frac{1}{2} \cos A - \frac{1}{4} \cos(A - 2\varphi) - \frac{1}{4} \cos(A + 2\varphi) - \frac{1}{2} \cos(A - 2u) + \frac{1}{4} \cos(2\varphi + 2u - A) + \frac{1}{4} \cos(A + 2\varphi - 2u)$
$-\cos A$	$-\cos A$
$+\cos(\varphi + u) \cos(A + \varphi - u)$	$\frac{1}{2} \cos(A + 2\varphi) + \frac{1}{2} \cos(A - 2u)$

Summing each column of Fourier modes gives 0 (e.g. the $\cos A$ modes: $\frac{1}{4} + \frac{1}{4} + \frac{1}{2} - 1 = 0$; the $\cos(A + 2\varphi)$ modes: $-\frac{1}{4} - \frac{1}{4} + \frac{1}{2} = 0$; the $\cos(A - 2u)$ modes: $\frac{1}{4} - \frac{1}{4} - \frac{1}{2} + \frac{1}{2} = 0$; the $\cos(2\varphi + 2u - A)$ modes: $-\frac{1}{4} + \frac{1}{4} = 0$; the $\cos(A - 2\varphi)$ modes: $\frac{1}{4} - \frac{1}{4} = 0$; the $\cos(A + 2\varphi - 2u)$ modes: $-\frac{1}{4} + \frac{1}{4} = 0$).
(bracket identity)

Therefore, defining

$$P_u := \cos A - \cos(\varphi + u) \cos(A + \varphi - u), \quad Q_u := \sin^2(\varphi + u),$$

equation (5) becomes the **K-side relation**

$$\boxed{\ell_K := b Q_u - c P_u = 0.} \quad (K)$$

L-side. The L-side computation is *literally the same computation* with the substitutions $u \rightarrow v$, $\chi \rightarrow \psi$, $b \leftrightarrow c$: (E4) has the same shape as (E3), and equating (E2) with (E6) has the same shape as equating (E1) with (E5) with b and c exchanged. Hence, with

$$P_v := \cos A - \cos(\varphi + v) \cos(A + \varphi - v), \quad Q_v := \sin^2(\varphi + v),$$

$$\boxed{\ell_L := c Q_v - b P_v = 0.} \quad (L)$$

Part 4. A, K, L are not collinear

Suppose, for contradiction, that A, K, L are collinear. Since $K, L \in \text{int}(ABC)$ and $K, L \neq A$, both lie on one ray ℓ from A through the interior of the triangle; write $\ell = \{A + t\hat{\ell} : t > 0\}$ and $K = A + t_K\hat{\ell}$, $L = A + t_L\hat{\ell}$ with $t_K, t_L > 0$.

Monotonicity claim. For $X(t) = A + t\hat{\ell}$ ($t > 0$), the function $t \mapsto \angle ABX(t)$ is strictly increasing, and so is $t \mapsto \angle ACX(t)$.

Proof. Take coordinates with A at the origin and $B = (c, 0)$; since ℓ enters the interior, $\hat{\ell} = (\cos \theta, \sin \theta)$ with $\sin \theta > 0$ (orient the C -side up), where $\theta \in (0, A)$. The angle at B between ray BA (direction $(-1, 0)$) and ray BX (direction $X(t) - B = (t \cos \theta - c, t \sin \theta)$) has

$$\cos \angle ABX = \frac{c - t \cos \theta}{|X - B|}, \quad \sin \angle ABX = \frac{t \sin \theta}{|X - B|} > 0,$$

so

$$\cot \angle ABX = \frac{c - t \cos \theta}{t \sin \theta} = \frac{1}{\sin \theta} \left(\frac{c}{t} - \cos \theta \right),$$

which is strictly decreasing in t ; since $\cot$ is strictly decreasing on $(0, \pi)$, $\angle ABX(t)$ is strictly increasing. The statement at C is the same computation in coordinates with A at the origin and C on the positive axis of the frame: there $\hat{\ell}$ makes angle $\theta' = A - \theta \in (0, A)$ with ray AC , $\sin \theta' > 0$, and $\cot \angle ACX = \frac{1}{\sin \theta'} \left(\frac{b}{t} - \cos \theta' \right)$, strictly decreasing.

Now, by (0d) and (0b), $\angle ABL = \varphi + \psi > \varphi = \angle ABK$ (as $\psi > 0$), so by the claim applied at B : $t_L > t_K$. But also $\angle ACK = \varphi + \chi > \varphi = \angle ACL$ (as $\chi > 0$), so by the claim applied at C : $t_K > t_L$. Contradiction. Hence A, K, L are not collinear, and the circumcircle ω and circumcentre O of triangle AKL exist and are unique.

Consequence. The rays AK and AL are distinct. Ray AK makes angle u with ray AB ; ray AL makes angle $A - v$ with ray AB ; both $u, A - v \in (0, A)$. Distinctness means $u \neq A - v$; since $|(A - v) - u| < A < \pi$,

$$\sin(A - u - v) \neq 0. \quad (6)$$

Part 5. Coordinates, the circumcentre, and the goal polynomial G

Place $A = (0, 0)$, $B = (c, 0)$, $C = (b \cos A, b \sin A)$ (so C is in the upper half plane; K and L , being interior, are as well). By Part 0,

$$K = AK(\cos u, \sin u), \quad L = AL(\cos(A - v), \sin(A - v)).$$

Since $A = 0 \in \omega$, the circle ω has equation $|z|^2 = 2\langle O, z \rangle$ ($z \in \mathbb{R}^2$), and for any point P ,

$$\text{pow}(P, \omega) = |P - O|^2 - |O|^2 = |P|^2 - 2\langle O, P \rangle.$$

With $R_\omega = OA = |O|$, we get $OB^2 - OC^2 = \text{pow}(B, \omega) - \text{pow}(C, \omega) = (c^2 - b^2) - 2\langle O, B - C \rangle$. By (1),

$$OM = ON \iff 2\langle O, B - C \rangle = \frac{c^2 - b^2}{2}. \quad (7)$$

Write $(X, Y) := 2O$. Since $K, L \in \omega$: $|K|^2 = 2\langle O, K \rangle$ and $|L|^2 = 2\langle O, L \rangle$; dividing by $AK > 0$ resp. $AL > 0$:

$$X \cos u + Y \sin u = AK, \quad X \cos(A - v) + Y \sin(A - v) = AL. \quad (8)$$

The determinant of this linear system is $\cos u \sin(A - v) - \sin u \cos(A - v) = \sin(A - u - v) \neq 0$ by (6), so Cramer's rule gives

$$X = \frac{AK \sin(A - v) - AL \sin u}{\sin(A - u - v)}, \quad Y = \frac{AL \cos u - AK \cos(A - v)}{\sin(A - u - v)}.$$

Since $B - C = (c - b \cos A, -b \sin A)$, condition (7) reads $X(c - b \cos A) - Y b \sin A = \frac{c^2 - b^2}{2}$, i.e.

$$AK[(c - b \cos A) \sin(A - v) + b \sin A \cos(A - v)] - AL[(c - b \cos A) \sin u + b \sin A \cos u] = \frac{c^2 - b^2}{2} \sin(A - u - v).$$

Using $\sin A \cos(A - v) - \cos A \sin(A - v) = \sin v$ and $\sin A \cos u - \cos A \sin u = \sin(A - u)$, the brackets simplify:

$$(c - b \cos A) \sin(A - v) + b \sin A \cos(A - v) = c \sin(A - v) + b \sin v,$$

$$(c - b \cos A) \sin u + b \sin A \cos u = c \sin u + b \sin(A - u).$$

So (7) is equivalent to

$$AK [c \sin(A - v) + b \sin v] - AL [c \sin u + b \sin(A - u)] = \frac{c^2 - b^2}{2} \sin(A - u - v). \quad (G0)$$

Finally substitute $AK = \frac{c \sin \varphi}{\sin(\varphi + u)}$, $AL = \frac{b \sin \varphi}{\sin(\varphi + v)}$ from (E1), (E2) and multiply by $\sin(\varphi + u) \sin(\varphi + v) > 0$: **(G0)**, hence $OM = ON$, is equivalent to $G = 0$, where

$$G := c \sin \varphi \sin(\varphi + v) [c \sin(A - v) + b \sin v] - b \sin \varphi \sin(\varphi + u) [c \sin u + b \sin(A - u)] - \frac{c^2 - b^2}{2} \sin(A - u - v) \sin(\varphi + u) \sin(\varphi + v).$$

Part 6. The certificate

Claim (certificate identity). With $P_u, Q_u, P_v, Q_v, \ell_K, \ell_L$ as in Part 3 and G as in (9), the following holds **identically in the six free real variables** u, v, φ, A, b, c (no constraints assumed):

$$2 \sin A \cdot G = \alpha \ell_K + \beta \ell_L, \quad (10)$$

where

$$\alpha := - \left[b (\sin^2 \varphi + \sin^2 v) + c \sin(\varphi + v) \sin(A - \varphi - v) \right],$$

$$\beta := c (\sin^2 \varphi + \sin^2 u) + b \sin(\varphi + u) \sin(A - \varphi - u).$$

Why the claim finishes the proof. By Part 3, the hypotheses force $\ell_K = \ell_L = 0$. Hence $2 \sin A \cdot G = 0$; since $A \in (0, \pi)$ is an angle of triangle ABC , $\sin A \neq 0$, so $G = 0$; by Part 5 this is equivalent to (7), and by Part 1 to $OM = ON$. ■ (modulo the claim)

Proof of the claim. Both sides of (10) are quadratic forms in (b, c) whose coefficients are trigonometric polynomials in (u, v, φ, A) ; it suffices to match the coefficients of b^2 , bc , c^2 . Collecting them from (9) and from $\alpha\ell_K + \beta\ell_L$ (recall $\ell_K = bQ_u - cP_u$, $\ell_L = cQ_v - bP_v$), the claim is equivalent to the three identities:

$$(\text{Id-}b^2) \quad 2 \sin A \left[-\sin \varphi \sin(\varphi+u) \sin(A-u) + \frac{1}{2} \sin(A-u-v) \sin(\varphi+u) \sin(\varphi+v) \right] = -(\sin^2 \varphi + \sin^2 v) \sin^2(\varphi+u) - \sin(\varphi+u) \sin(A-\varphi-u) P_v.$$

$$(\text{Id-}bc) \quad 2 \sin A \sin \varphi [\sin v \sin(\varphi+v) - \sin u \sin(\varphi+u)] = (\sin^2 \varphi + \sin^2 v) P_u - \sin(\varphi+v) \sin(A-\varphi-v) \sin^2(\varphi+u) + \sin(\varphi+u) \sin(A-\varphi-u) \sin^2(\varphi+v) - (\sin^2 \varphi + \sin^2 u) P_v.$$

$$(\text{Id-}c^2) \quad 2 \sin A \left[\sin \varphi \sin(\varphi+v) \sin(A-v) - \frac{1}{2} \sin(A-u-v) \sin(\varphi+u) \sin(\varphi+v) \right] = \sin(\varphi+v) \sin(A-\varphi-v) P_u + (\sin^2 \varphi + \sin^2 u) \sin^2(\varphi+v).$$

(Bookkeeping check for the coefficient collection: e.g. the b^2 -terms of $\alpha\ell_K$ come from the b -part of α times the b -part of ℓ_K : $-(\sin^2 \varphi + \sin^2 v) \cdot Q_u$; those of $\beta\ell_L$ from $b \sin(\varphi+u) \sin(A-\varphi-u) \cdot (-b P_v)$; similarly for bc and c^2 .)

(Id- c^2) follows from (Id- b^2). Swap $u \leftrightarrow v$ in (Id- b^2) and multiply by -1 : the left side becomes $2 \sin A [\sin \varphi \sin(\varphi+v) \sin(A-v) - \frac{1}{2} \sin(A-u-v) \sin(\varphi+u) \sin(\varphi+v)]$ (note $\sin(A-v-u) = \sin(A-u-v)$), and the right side becomes $(\sin^2 \varphi + \sin^2 u) \sin^2(\varphi+v) + \sin(\varphi+v) \sin(A-\varphi-v) P_u$ — exactly (Id- c^2). So only (Id- b^2) and (Id- bc) need verification.

Verification of (Id- b^2). Every term on both sides carries the factor $\sin(\varphi+u)$ (for the P_v -term this is explicit; for the left side, both bracket terms contain it). Dividing (Id- b^2) by it and moving everything to one side, it is equivalent to $T_1 + T_2 + T_3 + T_4 + T_5 = 0$ with the Fourier (product-to-sum) expansions:

term	Fourier expansion
$T_1 = \sin A \sin(\varphi+v) \sin(A-u-v)$	$\frac{1}{4} \sin(\varphi-u) - \frac{1}{4} \sin(2A+\varphi-u) +$
$T_2 = -2 \sin A \sin \varphi \sin(A-u)$	$\frac{1}{4} \sin(\varphi+u+2v) - \frac{1}{4} \sin(\varphi+u+2v-2A)$
$T_3 = (\sin^2 \varphi + \sin^2 v) \sin(\varphi+u)$	$-\frac{1}{2} \sin(\varphi-u) - \frac{1}{2} \sin(\varphi+u) + \frac{1}{2} \sin(\varphi+u-2A) +$
$T_4 = \sin(A-\varphi-u) \cos A$	$\frac{1}{2} \sin(2A+\varphi-u)$
$T_5 = -\sin(A-\varphi-u) \cos(\varphi+v) \cos(A+\varphi-v)$	$\frac{1}{4} \sin(\varphi-u) + \sin(\varphi+u) - \frac{1}{4} \sin(3\varphi+u) -$
	$\frac{1}{4} \sin(\varphi+u-2v) - \frac{1}{4} \sin(\varphi+u+2v)$
	$-\frac{1}{2} \sin(\varphi+u) - \frac{1}{2} \sin(\varphi+u-2A)$
	$\frac{1}{4} \sin(3\varphi+u) - \frac{1}{4} \sin(2A+\varphi-u) +$
	$\frac{1}{4} \sin(\varphi+u-2v) + \frac{1}{4} \sin(\varphi+u+2v-2A)$

(Here $T_4 + T_5 = \sin(A-\varphi-u) P_v$ after sign transport; each row is two applications of the standard product-to-sum formulas; note $\sin(\varphi+u-2A) = -\sin(2A-\varphi-u)$ etc., all rows written with a consistent sign convention $\sin(-X) = -\sin X$.)

Column-wise cancellation of the Fourier modes: $\sin(\varphi-u)$: $\frac{1}{4} - \frac{1}{2} + \frac{1}{4} = 0$; $\sin(2A+\varphi-u)$: $-\frac{1}{4} + \frac{1}{2} - \frac{1}{4} = 0$; $\sin(\varphi+u+2v)$: $\frac{1}{4} - \frac{1}{4} = 0$; $\sin(\varphi+u+2v-2A)$: $-\frac{1}{4} + \frac{1}{4} = 0$; $\sin(\varphi+u)$: $-\frac{1}{2} + 1 - \frac{1}{2} = 0$; $\sin(\varphi+u-2A)$: $\frac{1}{2} - \frac{1}{2} = 0$; $\sin(3\varphi+u)$: $-\frac{1}{4} + \frac{1}{4} = 0$; $\sin(\varphi+u-2v)$: $-\frac{1}{4} + \frac{1}{4} = 0$. All modes cancel, so (Id- b^2) holds.

Verification of (Id- bc). Moving everything to one side, (Id- bc) is equivalent to $S_1 + \dots + S_6 = 0$ with:

term	Fourier expansion
$S_1 = 2 \sin A \sin \varphi \sin v \sin(\varphi+v)$	$\frac{1}{4} \cos(A-2\varphi) - \frac{1}{4} \cos(A+2\varphi) + \frac{1}{4} \cos(A-2v) - \frac{1}{4} \cos(A+2v) - \frac{1}{4} \cos(2\varphi+2v-A) + \frac{1}{4} \cos(A+2\varphi+2v)$
$S_2 = -2 \sin A \sin \varphi \sin u \sin(\varphi+u)$	$-\frac{1}{4} \cos(A-2\varphi) + \frac{1}{4} \cos(A+2\varphi) - \frac{1}{4} \cos(A-2u) + \frac{1}{4} \cos(A+2u) + \frac{1}{4} \cos(2\varphi+2u-A) - \frac{1}{4} \cos(A+2\varphi+2u)$
$S_3 = -(\sin^2 \varphi + \sin^2 v) P_u$	$-\frac{9}{8} \cos A + \frac{1}{4} \cos(A-2\varphi) + \frac{3}{4} \cos(A+2\varphi) - \frac{1}{8} \cos(A+4\varphi) + \frac{1}{2} \cos(A-2u) + \frac{1}{4} \cos(A-2v) + \frac{1}{4} \cos(A+2v) - \frac{1}{8} \cos(2\varphi+2u-A) - \frac{1}{8} \cos(2u+2v-A) - \frac{1}{8} \cos(A+2\varphi-2u) - \frac{1}{8} \cos(A+2\varphi-2v) - \frac{1}{8} \cos(A+2\varphi+2v) - \frac{1}{8} \cos(A-2u+2v)$
$S_4 = \sin(\varphi+v) \sin(A-\varphi-v) \sin^2(\varphi+u)$	$-\frac{1}{4} \cos A + \frac{1}{8} \cos(2\varphi+2u-A) + \frac{1}{4} \cos(2\varphi+2v-A) + \frac{1}{8} \cos(A+2\varphi+2u) - \frac{1}{8} \cos(A+2u-2v) - \frac{1}{8} \cos(4\varphi+2u+2v-A)$
$S_5 = -\sin(\varphi+u) \sin(A-\varphi-u) \sin^2(\varphi+v)$	$\frac{1}{4} \cos A - \frac{1}{4} \cos(2\varphi+2u-A) - \frac{1}{8} \cos(2\varphi+2v-A) - \frac{1}{8} \cos(A+2\varphi+2v) + \frac{1}{8} \cos(A-2u+2v) + \frac{1}{8} \cos(4\varphi+2u+2v-A)$
$S_6 = (\sin^2 \varphi + \sin^2 u) P_v$	$\frac{9}{8} \cos A - \frac{1}{4} \cos(A-2\varphi) - \frac{3}{4} \cos(A+2\varphi) + \frac{1}{8} \cos(A+4\varphi) - \frac{1}{4} \cos(A-2u) - \frac{1}{4} \cos(A+2u) - \frac{1}{8} \cos(A-2v) + \frac{1}{8} \cos(2\varphi+2v-A) + \frac{1}{8} \cos(2u+2v-A) + \frac{1}{8} \cos(A+2\varphi-2u) + \frac{1}{8} \cos(A+2\varphi-2v) + \frac{1}{8} \cos(A+2\varphi+2u) + \frac{1}{8} \cos(A+2u-2v)$

Column-wise cancellation of all Fourier modes (each mode listed with its total): $\cos A$: $-\frac{9}{8} - \frac{1}{4} + \frac{1}{4} + \frac{9}{8} = 0$; $\cos(A-2\varphi)$: $\frac{1}{4} - \frac{1}{4} + \frac{1}{4} - \frac{1}{4} = 0$; $\cos(A+2\varphi)$: $-\frac{1}{4} + \frac{1}{4} + \frac{3}{4} - \frac{3}{4} = 0$; $\cos(A+4\varphi)$: $-\frac{1}{8} + \frac{1}{8} = 0$; $\cos(A-2u)$: $-\frac{1}{4} + \frac{1}{2} - \frac{1}{4} = 0$; $\cos(A+2u)$: $\frac{1}{4} - \frac{1}{4} = 0$; $\cos(A-2v)$: $\frac{1}{4} + \frac{1}{4} - \frac{1}{2} = 0$; $\cos(A+2v)$: $-\frac{1}{4} + \frac{1}{4} = 0$; $\cos(2\varphi+2u-A)$: $\frac{1}{4} - \frac{1}{8} + \frac{1}{8} - \frac{1}{4} = 0$; $\cos(2\varphi+2v-A)$: $-\frac{1}{4} + \frac{1}{4} - \frac{1}{8} + \frac{1}{8} = 0$; $\cos(A+2\varphi+2u)$: $-\frac{1}{4} + \frac{1}{8} + \frac{1}{8} = 0$; $\cos(A+2\varphi+2v)$: $\frac{1}{4} - \frac{1}{8} - \frac{1}{8} = 0$; $\cos(A+2\varphi-2u)$: $-\frac{1}{8} + \frac{1}{8} = 0$; $\cos(A+2\varphi-2v)$: $-\frac{1}{8} + \frac{1}{8} = 0$; $\cos(2u+2v-A)$: $-\frac{1}{8} + \frac{1}{8} = 0$; $\cos(A-2u+2v)$: $-\frac{1}{8} + \frac{1}{8} = 0$; $\cos(A+2u-2v)$: $-\frac{1}{8} + \frac{1}{8} = 0$; $\cos(4\varphi+2u+2v-A)$: $-\frac{1}{8} + \frac{1}{8} = 0$. All modes cancel, so (Id- bc) holds.

This proves the certificate identity (10), and with it the theorem:

$$\boxed{OM = ON.}$$

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